

II. Chemical Reaction of polymers

Reaction of synthetic polymers

Polymer can be modified to improve some properties such as compatibility, fire retardant or adhesion or to provide specific functional gr.

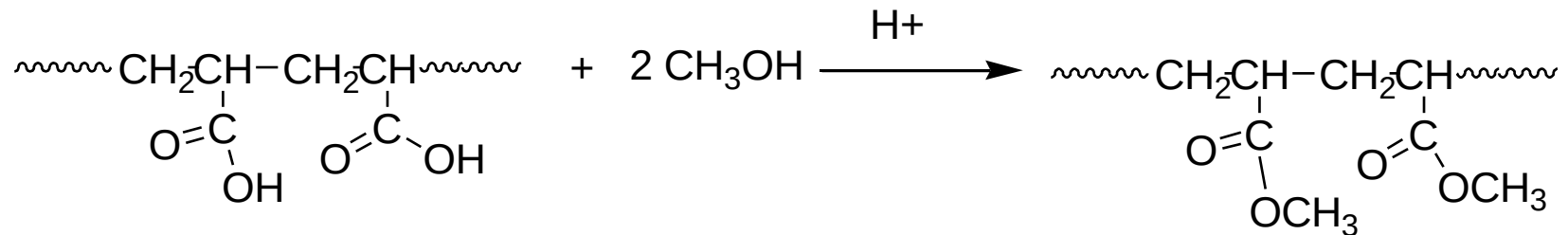
For example :

- Chlorinated of PP, PE, PVC to increase its softening temp. and compatibility with other polymers and decrease flammability.
- Ion exchange polystyrene.
- PVA preparation from PVAc
- So on...

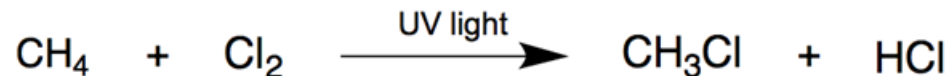
Nature of polymer reaction

Reaction of polymer depends on functional gr. even they are high mol. wt

Esterification:



Halogenation:



Nature of polymer reaction

The reaction of polymer quite slow compared with small molecular.

Factors affect to the chemical reactivity of polymer:

- ❑ Crystallinity
- ❑ Solubility
- ❑ Neighboring groups : steric effect or electrostatic effect.
- ❑ Conformation of polymer chain

I. Reaction of cellulose

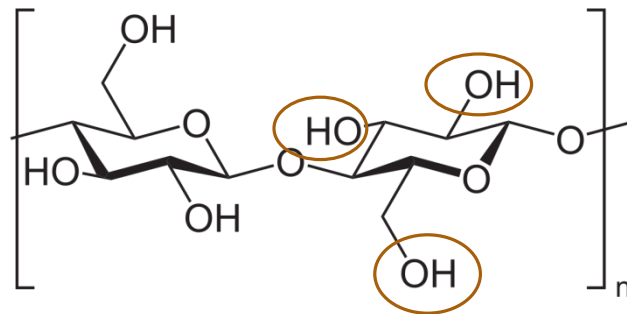
Cellulose from wood pulp and short fiber of cotton.

MW $\cong$ 300,000 – 1,000,000 g/mol.

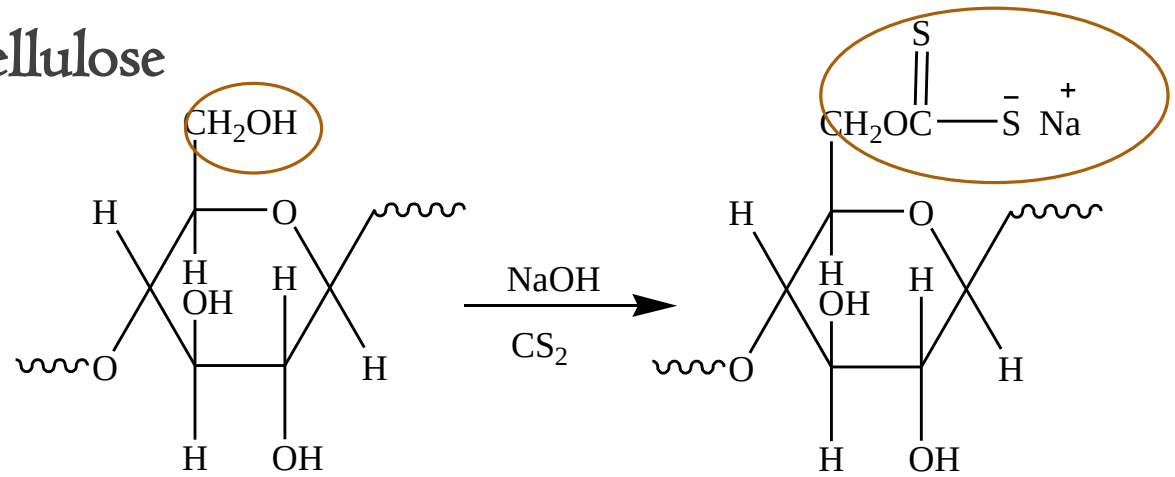
It consists of 3 –OH/unit. The rigid chain of cellulose is strongly H-bond and high crystalline.

It is insoluble and infusible (degrades before melt).

Cellulose film and fiber obtain by chemical modification of cellulose.



I.I Dissolution of cellulose

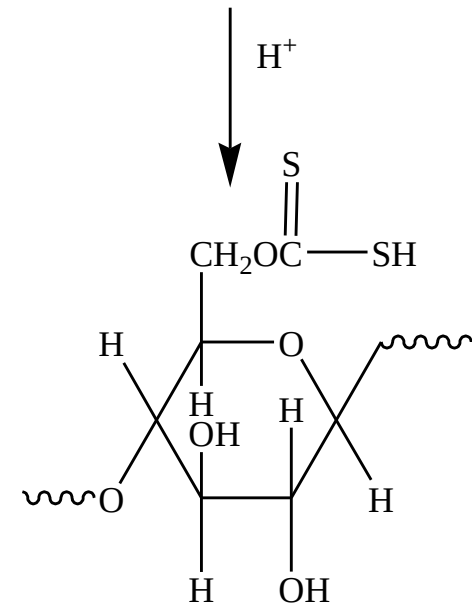


cellulose sodium xanthate

Cellulose fiber (rayon) or film (cellophane) obtained from *viscose* process.

1. Cellulose pulp is reacted with CS_2 to form cellulose xanthate (soluble in caustic solution).
2. Convert cellulose xanthate back to cellulose (called regeneration) by treatment with aq. H_2SO_4

$-\text{CS}_2$
 $-\text{H}_2\text{O}$

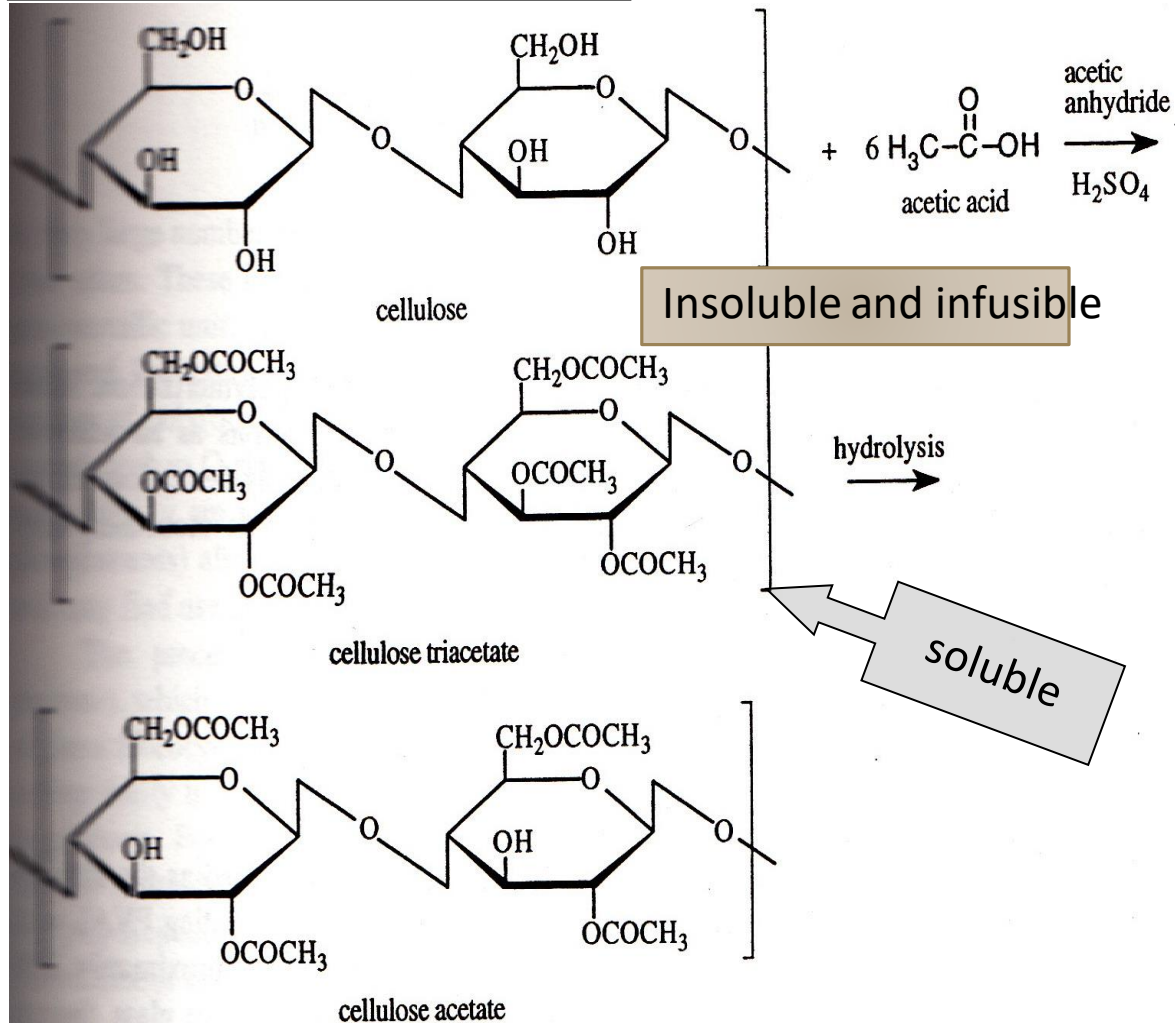


cellulose xanthic acid

“Regeneration process”

1.2 Esterification of cellulose

Partial acetylated cellulose

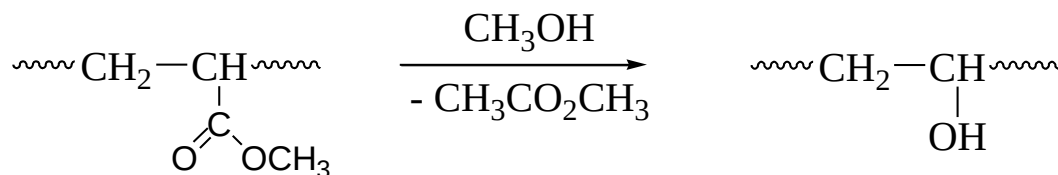


65-75% acetylated; Amorphous and highly soluble in solvent

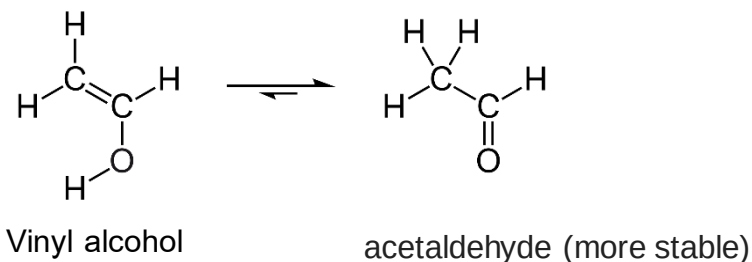
Cellulose acetate : reaction of cellulose and glacial acetic acid in presence of acetic anhydride and trace H_2SO_4 in refluxing CH_2Cl_2 . Cellulose triacetate can be wet spun into fiber ($\text{CH}_2\text{Cl}_2 + \text{AcI}$), then partial hydrolyzed to give cellulose acetate.

2. Reaction of poly(vinyl acetate)

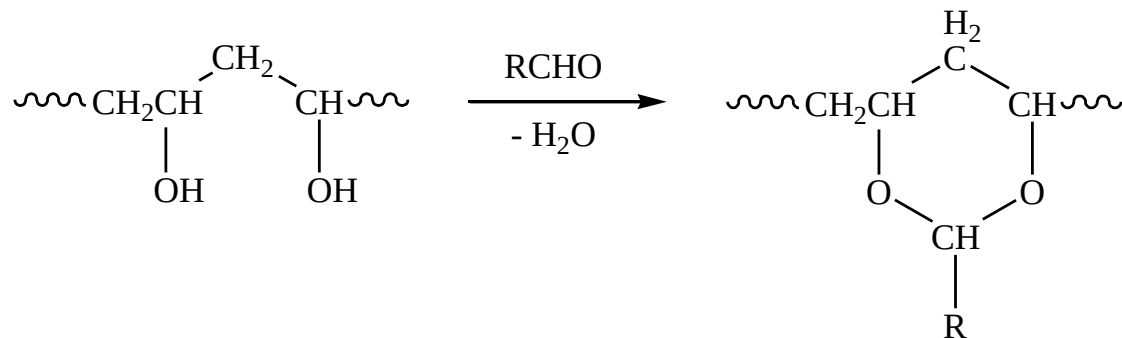
Poly(vinyl acetate), PVAc use as starting materials to synthesize Poly(vinyl alcohol) by alcoholysis with methanol in presence of acid or base catalyst.



Vinyl alcohol are unstable cpd.

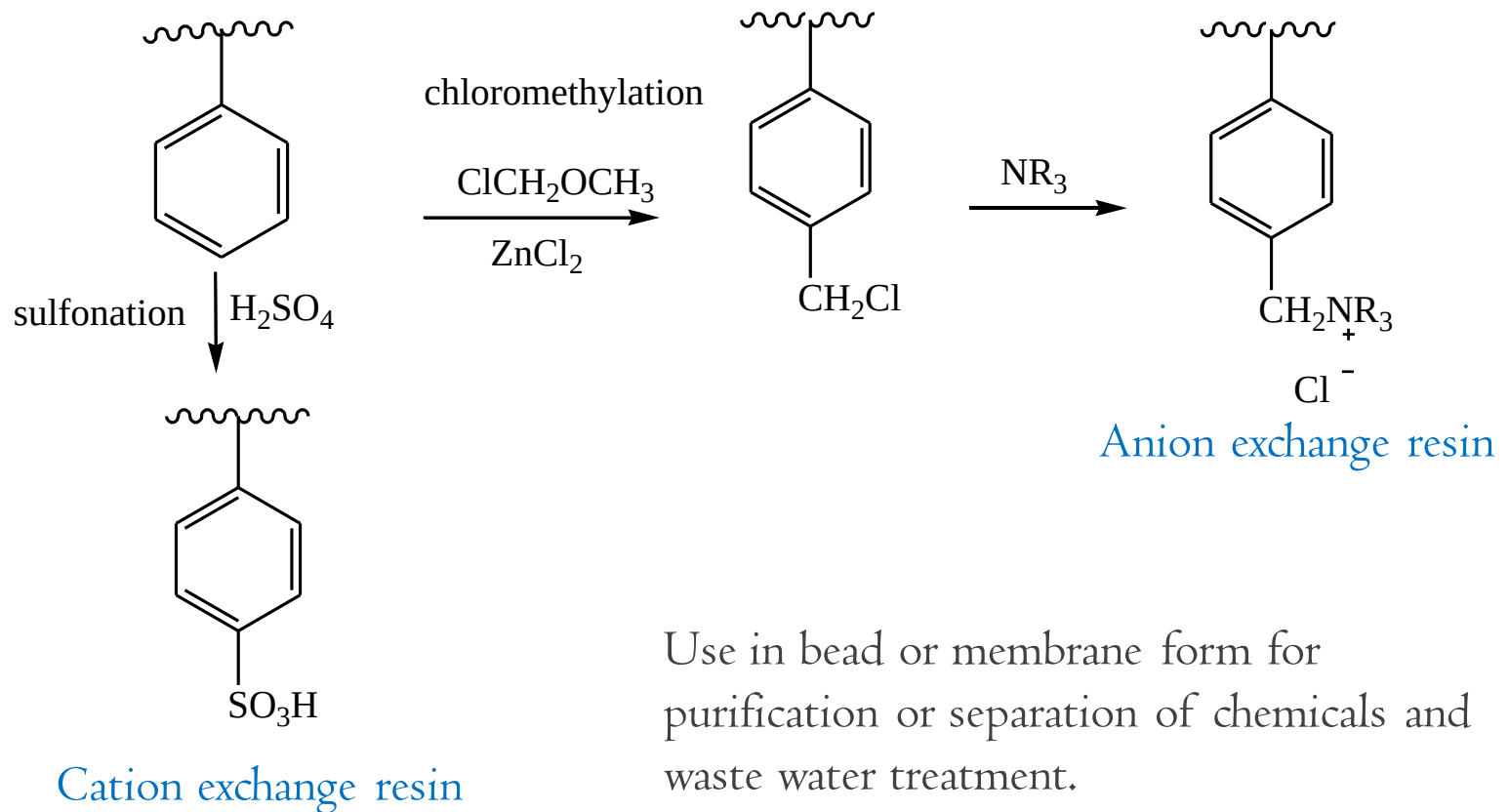


Reaction of PVA and aldehyde under acid catalyzed, get poly(vinyl acetal)



3. Aromatic substitution

Aromatic substitution of Polystyrene with sulfonic acid or quaternary ammonium in commercial ion exchange resin preparation

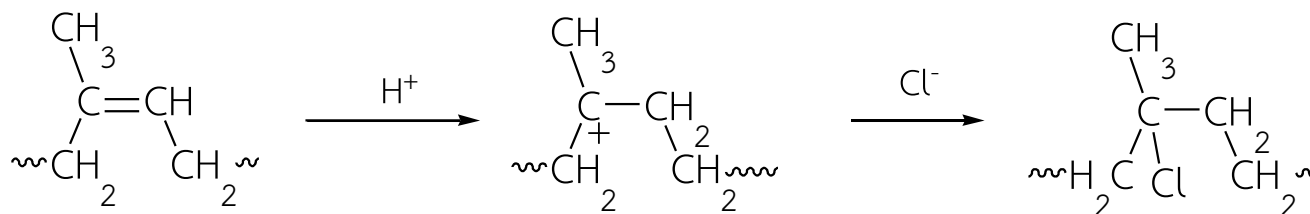


4. Halogenation

❖ Natural rubber

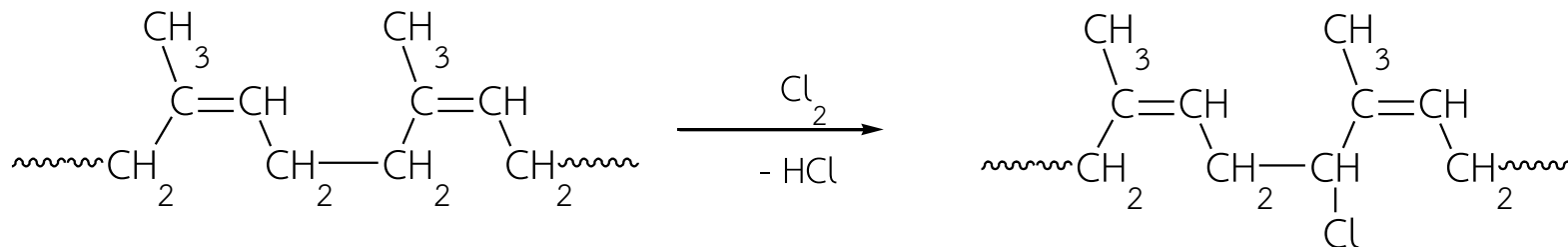
- Hydrochlorination

The mechanism is ionic reaction in CH_2Cl_2 , toluene, benzene



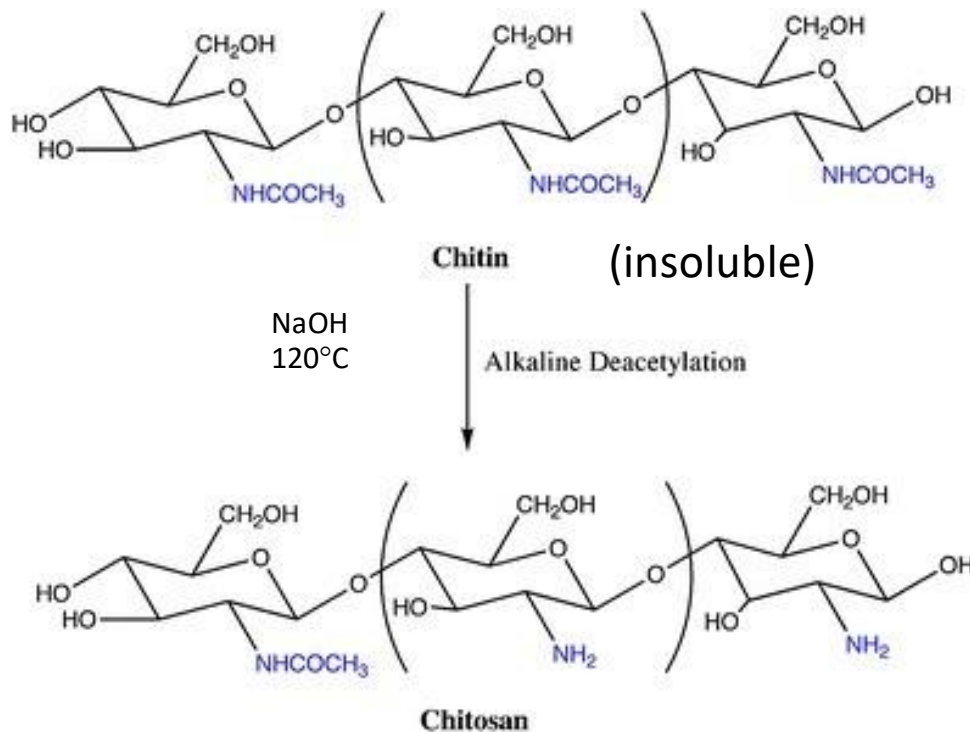
- Chlorination

The reaction is a free radical, Cl free radical substitute at allylic hydrogen.



5. Deacetylation of Chitin

Change some amide gr. of chitin to amine gr. in chitosan.



Soluble in organic acid

- Change chitin (insoluble in solv.) to chitosan via deacetylation of partial amide group to amine group which make chitosan is soluble in organic acid.
- Applications of chitosan: food additive, medical and pharmaceutical applications and so on.

References

1. ชัยวัฒน์ เจนวนิชย์, เคมีโพลิเมอร์พื้นฐาน, กรุงเทพฯ : โอเดียนสโตร์, 2527
2. ปรีชา พหลเทพ, โพลิเมอร์, พิมพ์ครั้งที่ 7, กรุงเทพฯ : มหาวิทยาลัยรามคำแหง, 2537
3. Billmeyer, Fred W., Text Book of Polymer Science, 3rded., John Wiley & Sons, 1984
4. Fried, J. R., Polymer Science and Technology, 2nded., Prentice Hall Professional Technical Reference, 2003
5. <http://www.kimica.jp/eng-chitosan-chemi-2.htm>